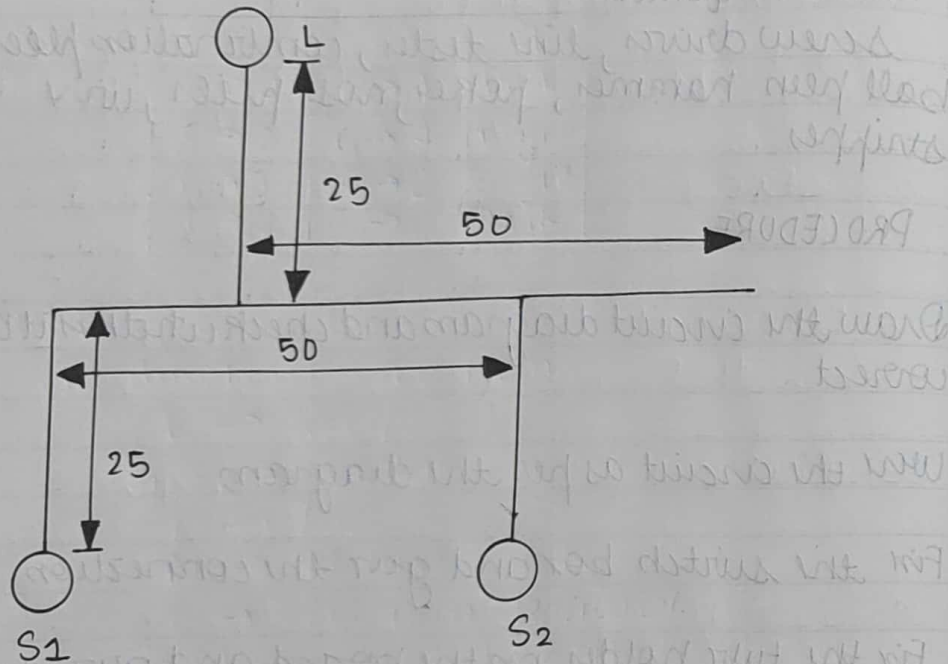
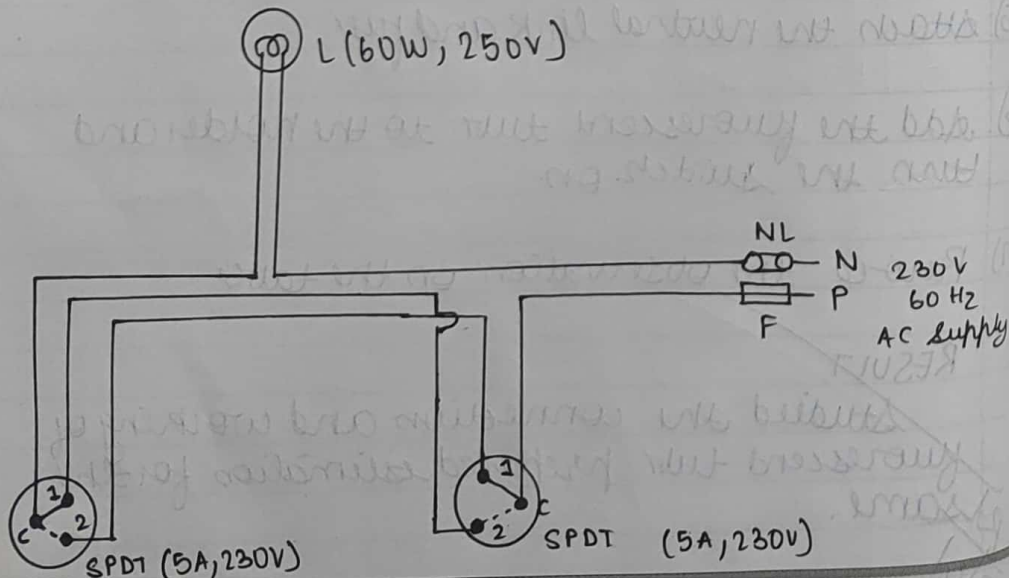


LAYOUT DIAGRAM



CIRCUIT DIAGRAM



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STAIRCASE WIRING

AIM

Wire up a circuit diagram to obtain one lamp controlled by two SPDT switches and prepare an estimation for the same

DESIGN

$$\text{Power} = P = VI \cos \phi$$

$$I = \frac{P}{V}$$

$$P = 60W \quad V = 230V$$

$$\text{Load Current} = I = \frac{60}{230} = \underline{\underline{0.26A}}$$

$$\text{Fuse wire rating} = 1.5 \times \text{Load Current}$$

$$= 1.5 \times 0.26A$$

$$= \underline{\underline{0.39A}}$$

OBSERVATION TABLE

S1	S2	BL
C → 1	C → 1	Dark
C → 2	C → 1	Bright
C → 1	C → 2	Bright
C → 2	C → 2	Dark

NOTE

all dimensions are in cm

L → Lamp

S₁, S₂ → Switch

F → Fuse

N → Neutral Link

C → Common terminal

SPDT → single pole double throw

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MATERIALS REQUIRED

S.No	Items	Specification	Rating	Quantity
1	Switch	SPDT (single pole double throw)	5A, 230V	2
2)	Lamp	Incandescent	60W, 250V	1
3)	Conduit	20mm PVC		175cm
4)	Copper wire	1 sq. mm		440cm
5)	Junction	20mm, 3way PVC		2
		20mm 2way 90° PVC		1
6)	Round Block	3 inch PVC		1
7)	Switch Block	3x2" PVC		2
8)	Clamp / Saddle	20mm metal		6
9)	Fuse unit	Kit Kat (Ordinary survivable)	16A	1
10)	Fuse wire	Tin & Lead Alloy	2A	10cm

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11.	Neutral link	Porcelain base with copper segment	16A	1
12.	Insulation Tape			1
13.	Screw	1/2"		12
		3/4"		6
		1 1/4"		1

TOOLS REQUIRED

- Screw driver, line tester, combination pliers, ball pen hammer, poker, nose pliers, wire strippers.

PROCEDURE

- Draw the circuit diagram and check whether it is correct
- Fix the conduit to the board using saddle of appropriate diameter
- Wire the circuit as per the circuit diagram
- Fix the switch box and give its connection
- Fix the round block to the base board

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6. Fix the lamp holder on top of the round block and give its connection
7. attach neutral link and fuse
8. Add the incandescent lamp to the holder and turn the switch on
9. Record observation on the observation table

RESULT

A circuit with one lamp controlled by SPDT switch was wired up and an estimation was prepared for the same.

